

Software Engineering Group

Project

COMP2002 / G52GRP: Interim Report

Project information:

Project title:

Data Driven Sustainability

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Industry Sponsor:

Eviden

Industry Supervisors:

Charlotte Fardell, Brooke Timms

Team information:

Team number: 39

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Jacob Stephens	20479737	psyas14	Deputy Team Leader
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Daniel Swinbank	20453919	psyds13	Deputy Git Master
Humais Ansari	20381671	psyha7	Git Helper
Dixon Okoro	20363242	psydo14	Git Helper

Documentation (Links):

Trello board: Have been using Git Issue Boards and Git Issues

Code repository: https://projects.cs.nott.ac.uk/comp2002/2023-2024/team39_project.git

Document repository: https://projects.cs.nott.ac.uk/comp2002/2023-2024/team39_project.git

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Section 1- Interim Report

Section 1 - Project Background & Understanding

Initial Brief Understanding:

The project centres around the exploration and enhancement of sustainable practices through the application of machine learning models or advanced analytics. The goal is to develop and showcase a Proof of Concept or final solution that highlights the optimisation or improvement of these practices.

The significance of this project can be seen in its contribution to both social and environmental sustainability, two critical elements that are also related to business growth. Embracing these aspects is essential, especially considering the global commitment to achieving net-zero carbon goals.

The team's vision for this endeavour involves the creation of a data-driven sustainable web application. This application will harness the potential of machine learning algorithms to forecast photovoltaic power consumption while calculating potential savings. The specifics of this vision are detailed below. The teams initial understanding of the brief can be seen in there EOI that they produced, as on the document itself, the team outlined a proposed approach to the project. Here it is:

Our Proposed Approach to Eviden's Data-Driven Sustainability Project:

- **Use Case Selection:** We propose selecting a use case that demonstrates significant environmental sustainability impact, such as optimizing data center energy usage using AI algorithms to reduce carbon emissions.
- **Platform Selection and User Interfaces:** We will carefully choose a platform (on-prem or cloud) based on accessibility, cost, and carbon footprint considerations. Intuitive dashboards will be developed to ensure user-friendly interactions.
- **AI Integration:** If applicable, we will integrate AI technologies for data analysis, pattern recognition, and predictive modeling, enhancing sustainability efforts.
- **Security and Access Control:** We will implement a robust authentication and authorization system, ensuring data encryption, secure APIs, and regular security audits to safeguard sensitive information.
- **Stakeholder Engagement and Project Planning:** Stakeholders will be actively involved throughout the project, and a detailed project plan, justified based on project complexity and team preferences, will be developed using tools like Gantt Charts.
- **Monitoring, Evaluation, and Documentation:** We will continuously collect relevant data to assess the solution's impact on sustainability metrics. Regular evaluation against set objectives, along with comprehensive documentation, will ensure knowledge retention and project success.

The Teams Interpretation, Additions, and Innovations

In line with this vision, one of the team's key contributions involves the development and delivery of an interactive final solution. This solution will serve as a demonstrative platform highlighting the advantages and benefits of sustainable energy generation through the application of machine learning models.

Furthermore, the team has set distinct timelines and milestones for the project. While external deadlines for the prototype and final solution exist, the team have established additional internal deadlines. These serve as motivational checkpoints to ensure steady progress and keep the project on track.

Depth and Context of the Problem

Measuring the depth and context of the problem involves evaluating its potential contribution to sustainable practices, crucial for understanding the project's positive impact on sustainability. Alongside this evaluation, the team decided that establishing quantifiable metrics that we could compare our project to as one of our main priorities. By project completion, they aim to directly calculate environmental impact metrics, focusing on efficiency improvements and cost savings. Furthermore, ethical and privacy considerations form vital components that will be addressed throughout the project. Additionally, recognising potential challenges, particularly highlighting the critical role of data quality, is paramount. The project's success hinges on adhering to industry-standard data quality and completeness when dealing with large datasets to ensure the accuracy of the team's machine learning models. Finding and mitigating these challenges is pivotal for achieving the overall project goals.

The Team's Current Solutions:

Through weekly sprints and meetings, the project has resulted in very comprehensive deliverables which cover all the different areas of sustainable data-driven solutions. Commencing with detailed research and design phases, the team created in-depth reports on use cases, data flow diagrams, and initial UI designs alongside demo videos, all meticulously documented and refined through iterative feedback.

After that phase, the team split into groups and started the development of backend infrastructure, UI prototypes, and AI-driven features. These teams each produced a working prototype web app, a database to store all the data used by the AI, and evidence of AI model implementation. Rigorous testing and optimisation led to meticulous security integration, UI testing reports, and user manuals, ensuring robustness and usability.

Section 2 - Requirements & Critical Analysis

The team's approach to requirements gathering was simple yet effective. The team made sure that they broke down the Project Specification document into more manageable pieces and created a plan on how they would meet these requirements as a team. This was important in the preliminary stages of development as everyone needed to have a good understanding of what the final product should look like and how they plan to get there.

To make sure that everyone on the team was clear on the requirements and general approach, Andreas created a 'Project Evaluation Document.' This document contained a general step by step guide that the team could take to produce an adequate result. The openness in the industry requirements gave the team the freedom to shape the project but also faced the task of defining clear, actionable objectives. This required team deliberation and iterative feedback from Charlotte and the academic supervisor. Adopting a top-down approach, the team managed to break down overarching theme of 'sustainability' into tangible, more specific requirements. Here are some screenshots of this:

Eviden - Data - Driven Sustainability:

- Overview of the Project:

- The objective is to investigate and recommend cutting-edge best practices, conduct comprehensive impact analyses of emerging technologies in the environmentally-conscious IT landscape, and deliver a Proof of Concept (POC). The focus is on transforming existing processes, developing sustainable business models, and coexisting with emerging technologies through data-driven decision-making and AI integration

- Benefits of Eviden:

- **Expertise:** Eviden specializes in data-driven sustainability solutions, ensuring in-depth knowledge and experience in the field.
- **Innovation:** Eviden employs innovative approaches, including AI and advanced data analytics, fostering creative solutions for sustainability challenges.
- **Customization:** Eviden tailors solutions to specific business needs, ensuring relevance and effectiveness in achieving sustainability goals.
- **Eco-Conscious Approach:** Eviden considers the carbon footprint of technology choices, promoting environmentally responsible decision-making.

- Suggested Approach:

1) Selecting a Use Case for Environmental Sustainability:

- Choose a use case that demonstrates environmental sustainability value, showcasing the impact on sustainability challenges.
- Example - Optimize data center energy usage using AI algorithms, reducing carbon emissions significantly.

2) Developing the Proof of Concept (POC) or Final Solution:

- Platform Selection - Choose an appropriate platform (on-prem or cloud) based on accessibility, cost, and carbon footprint considerations.
- User Interfaces - Develop intuitive dashboards for end-users to interact with the 'data-driven sustainable' application, ensuring user-friendliness.
- Incorporate AI - If applicable, integrate AI technologies for data analysis, pattern recognition, and predictive modeling to enhance sustainability efforts.

3) Ensuring Security and Access Control:

- Implement a robust authentication and authorization system, differentiating between end-users and administrators for data management and security purposes.
- Ensure data encryption, secure APIs, and regular security audits to safeguard sensitive information.

4) Stakeholder Engagement and Project Planning:

- Involve stakeholders throughout the project to gather feedback, ensuring alignment with business goals.
- Project Plan - Develop a detailed project plan, utilizing tools like Gantt Charts. Justify the chosen method and format based on project complexity and team preferences.
- Measurable Objectives: Define specific, measurable objectives aligned with sustainability goals, allowing progress tracking and accountability.

5) Monitoring and Evaluation:

- Continuously collect relevant data to assess the impact of the implemented solution on sustainability metrics.
- Regularly evaluate the solution's performance against set objectives, making necessary adjustments to optimize outcomes.

6) Documentation and Knowledge Sharing:

- Maintain comprehensive documentation, including codebase, system architecture, and user manuals, ensuring knowledge retention.

One way of making sure that the team was working with a tailored set of requirements was through weekly meetings with the company contact, Charlotte Fardell. As the project had a lack of a conventional user base, the team had to rely on the user feedback given to us by Charlotte. With this feedback, they were able to take an adaptive approach to requirements gathering for the product they wanted to produce, which meant they had to be ready for changes at any point to the project.

In the following sprint after the team meetings with Charlotte, all teams prioritised incorporating Charlotte's feedback and suggestions, making sure that they produce the best result possible and that what they are producing aligns with Eviden's criteria.

Without direct user input like focus groups or surveys, the team made a point of meeting twice per sprint. These gatherings were all about understanding where the project was headed. This was all decided within team discussions, trying to see things from different perspectives. These sessions were team's way of getting close to what user feedback might offer. By dissecting and talking through everything thoroughly, they aimed to stay on track with what the industry sponsor wanted and where they wanted the project to go.

As for the technologies and tools the team used, their primary communication platform was Microsoft Teams, being instrumental in organising and documenting discussions, decisions, and individual feedback. It allowed us to keep a clear and traceable record of the evolving project requirements.

On the UI side of the project, the team used Figma's interactive prototyping capabilities to create a high-level mock-up of what the team's web app should look like. This enabled the team to visualise the team's concepts and present them to Charlotte for feedback. This tool was particularly useful in rapidly iterating design elements based on the sponsor's input. As for the actual development of the web app, the team used React and Node.js as it allowed for modularity in the code base. As React has a Component-Based Architecture, it means that components encapsulate their logic and UI, making development modular and easier.

The team's usage of Git was key in managing different versions of the project. It allowed for smooth collaboration among team members, allowing us to track changes, merge different branches, and keep a coherent codebase. For the AI sub-team, Jupyter Notebooks supplied an interactive environment to develop and test machine learning models, crucial for the project's data-driven approach. They employed Amazon RDS to manage the project's database needs as it offered scalability, security, and ease of integration, which were essential for the effective handling of data in the team's project.

Section 3 - Initial Software implementation & Testing

Data Management in Software Development

Data Storage and Retrieval Techniques

For our database architecture, we implemented Amazon Relational Database Service (Amazon RDS), which plays a crucial role in ensuring efficient interactions. Our choice of Amazon Relational Database Service (Amazon RDS) for integrating with the AI Evaluation Model is central to our project's success. We selected RDS for its scalable nature which is essential for managing the increasing data demands as our AI models evolve. This scalability ensures maintained performance levels despite growing system requirements.

Key to RDS's utility is its automated maintenance and vigorous security features. These handle crucial tasks like hardware provisioning, database setup, patching, and backups, therefore reducing our administrative workload and allowing focus on more innovative aspects of the project. Additionally, security measures such as data encryption and network isolation safeguard all data, aligning with stringent security standards.

The integration with our AI Evaluation Model is vital, involving organised storage of diverse data, from user inputs to AI-analysed results. This organisation enables efficient and safe data evaluation. Post-processing, the AI updates the database with fresh information, maintaining a dynamic and up-to-date data flow, crucial for precise data analysis.

Data access and retrieval in our system are streamlined through structured, relational formatting using MySQL. This approach is fundamental for effective and secure database management. MySQL not only aids in efficient data population but also ensures swift data access, vital for the AI Model's timely operation. Enhancing this, key field indexing reduces query times and boosts system responsiveness. Our comprehensive data management practices, including regular backups and data structure reviews, further assure continued data quality and relevance.

Handling Personal Data

Our system adheres to strict data security standards and complies with data protection regulations, particularly the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018. One of our key compliance procedures is data minimisation. We collect only the necessary data, reducing the risks of data breaches and aligning with UKGDPR principles. User consent is another fundamental principle applied transparently in our system's architecture. This approach allows users to provide consent for their data use and to review their data within the system. All these aspects contribute to a secure and safe data environment, enhancing the overall trust of our system.

The user interface of our system, as depicted in Figure 1, is carefully designed to enhance data privacy. A consent popup is prominently displayed to users, providing clear information about how their data is used within the system. Additionally, Figure 2 demonstrates our system's data management functionality. This feature allows users to actively manage their personal data, including reviewing, editing, or removing data as needed. This functionality ensures data accuracy and gives users direct control over their data, enhancing the overall user experience and trust in the system.

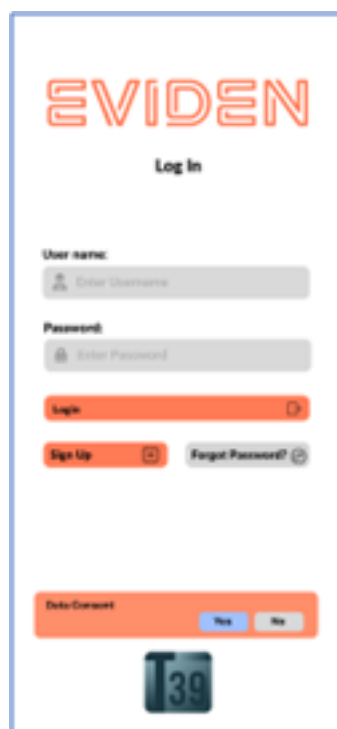


Figure 1



Figure 2

Software Architecture and Prototype Design

Architectural Overview

Our system's architecture, centred around the AI Evaluation Model, processes client inputs and environmental data for attuned recommendations. The user interface, acting as the first point of contact, collects specific data like energy usage and location. The AI then evaluates these inputs for feedback. Data flow is carefully managed, channelling inputs to the AI and storing processed results in a secure Amazon RDS cloud database, ensuring integrity and confidentiality. This design offers a modular, scalable framework for future enhancements.



Figure 3

As shown in Figure 3, the user interface is designed with a visually appealing orange theme to align with the client's company logo and aesthetic. The interface allows users to input their credentials and navigate through various tabs. The primary function of the software is to evaluate environmental factors using AI, offering users tailored proclamations based on their specified use case.

Our system utilises Microservices Architecture for independent development of components like the User Interface, AI Evaluation Model, and Data Storage, enhancing scalability and component independence. Additionally, the Publish-Subscribe pattern manages AI-data interactions, promoting efficient, separate data flow. These architectural patterns collectively ensure a strong, adaptable system, suitable for current operations and future expansions.

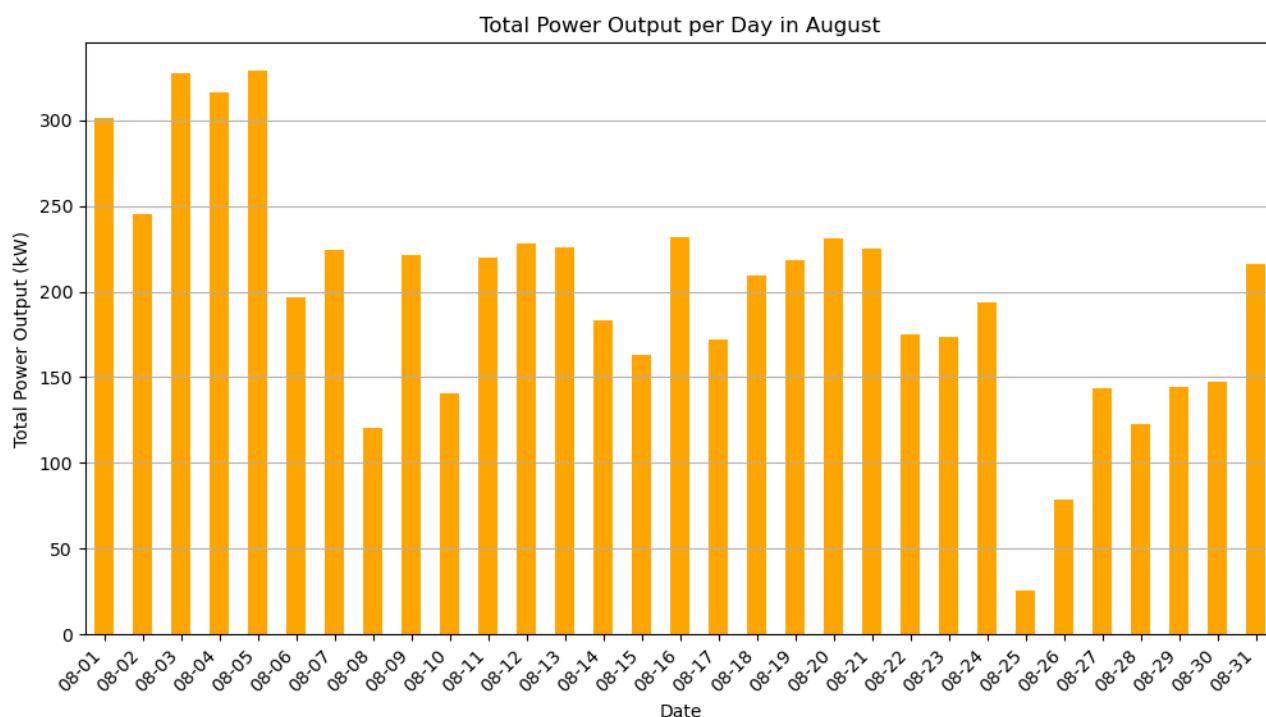
In addition, we implemented an Event-Driven type Architecture. This means that the UI elements communicate through events, so an action or change in one element triggers an event in another element. This allowed for a dynamic and reactive interface, but also for a web app that facilitates real-time updates.

Design Considerations

For the user interface, our strategy prioritised ease-of-use and accessibility within a web-based application. This approach allows for smooth interaction with various devices and audiences. The UI's intuitive layout simplifies navigation and data input processes, enhancing the overall user experience.

Proof of Concept

In the early development stages, we created various prototype models to showcase the key functionalities and project trajectory. The UI prototype, represented in Figure 3, offers a basic yet functional layout for user data input and result viewing based on the chosen model. This prototype was crucial for understanding the client's needs and gathering valuable feedback. Additionally, we developed an entry-level AI Linear Regression Model that promotes potential company energy savings from converting to sustainable energy sources, as demonstrated in Figure 4. This prototype offers a glimpse into the system's potential to provide constructive and customisable feedback.



Version Control and Code Management

Version Control Strategies

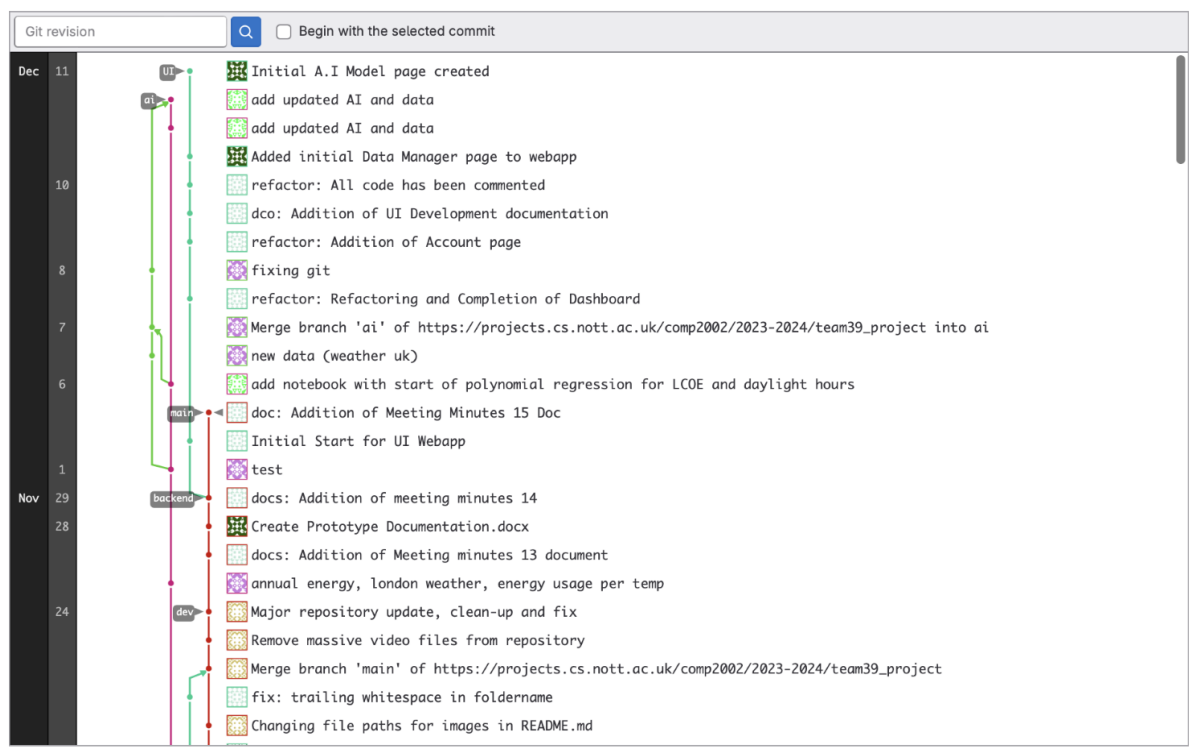
Team 39 has implemented various branching and merging techniques, with an emphasis on feature branching. This led to the creation of distinct branches for each specialised sub-division, including AI, Web Development, and Database. Each team operates on its dedicated branch, focusing on uninterrupted development.

In our version control process, we adhered to best practices to ensure organisation and code quality. Regular commits with descriptive messages provide a clear track of changes and enhance team comprehension. We maintain a clean and organised Git repository, avoiding large commits. Strategic branching separates distinctive features, preserving the stability of the main branch. Consistent review and merging of branches prevent divergence, additionally, regular backups safeguard our codebase.

Practical Applications

Branching and merging have been procedurally fundamental in our development. Separate branches for AI, backend, and UI allow each sub-division to work coherently without creating a fragile codebase. Team members conduct code reviews to ensure smooth integration, as shown in Figure 5.

Active branches			
UI	6ef3f36f · Initial A.I Model page created · 13 hours ago	17	New Download
ai	37e913f3 · add updated AI and data · 17 hours ago	12 8	New Download
main	9caeee4d · doc: Addition of Meeting Minutes 15 Doc · 5 days ago		Download
backend	c3755e58 · docs: Addition of meeting minutes 14 · 1 week ago	1 0	New Download
dev	a05d7094 · Major repository update, clean-up and fix · 2 weeks ago	4 0	New Download



Quality Assurance through Testing

Testing Methodologies

Throughout development, we engaged in rigorous testing methodologies. Unit testing with JavaScript frameworks like Jest ensures thorough tests on each class for proper functionality. Integration testing identifies flaws within each unit, and user acceptance testing confirms compliance with user expectations. These methods ensure code quality and efficiency. Unit testing isolates small components, allowing for early bug fixes. Integration testing ensures component compatibility, and user acceptance testing validates end-to-end functionality and user experience.

As shown in Figure 5, our testing tables provide thorough examples of component stress testing, ensuring code integrity and quality. Components such as the App, Dashboard, and Login are continuously tested for data quality and integrity. Testing has been vital in our development process, providing feedback and insights at various stages and ensuring a solid approach moving forward.

Testing Table:									
Component/File	Functionality Tested	Testing Method	Expected Outcome	Test Result		Sign up and Forget Password buttons functionality	Click simulation	-	Not Tested
LoadingScreenComponent.js	Rendering loading screen	Visual inspection	Loading screen displayed with Eviden logo, slogan and team image	Pass	DashboardComponent.js	Rendering of dashboard components and navigation	Visual inspection	Dashboard displayed with navigation links and content	Pass
	Navigation to login on click	Click simulation	Redirects to the 'login' page	Pass		Log out functionality	Click simulation	Removes login status and redirects to '/'	Pass
LoginComponent.js	Rendering login form	Visual inspection	Login form with fields for username, password, and buttons	Pass	AccountPage.js	Rendering of account settings components and navigation	Visual inspection	Account settings displayed with navigation links and content	Pass
	Login functionality	Simulate valid login	Successful login redirects to 'dashboard'	Pass		Change Name, Email, Password, Profile functionality	Click simulation	-	Not Tested
	Login functionality	Simulate invalid login	Invalid alert pops up saying the invalid user or password was inputted, reinput	Pass					

App.js	Routing setup	Inspection of routes and navigation	Routes '/login', '/dashboard', '/account' working correctly	Pass
ProtectedRouteComponent.js	Handling protected routes based on login status	Inspection of routing logic	'/dashboard' accessible when logged in, else redirects to '/login'	Pass

Figure 5

Documentation and Code Readability

Documentation Approach

In-line documentation examples show our approach to clarity and assistance for team members involved in reviewing, developing, and testing. This documentation ensures a clear understanding of the code and aids in future development and maintenance.

```

// Importing React library to use React features and importing hook for navigation
import React from 'react';
import { useNavigate } from 'react-router-dom';

// Define a functional component called AccountPage
const AccountPage = () => {
  const navigate = useNavigate();

  // Function to handle signing out
  const handleSignOut = () => {
    // Remove the isLoggedIn key from localStorage
    localStorage.removeItem('isLoggedIn');
    // Navigate the user to the loading page '/'
    navigate('/');
  };
};

```

```

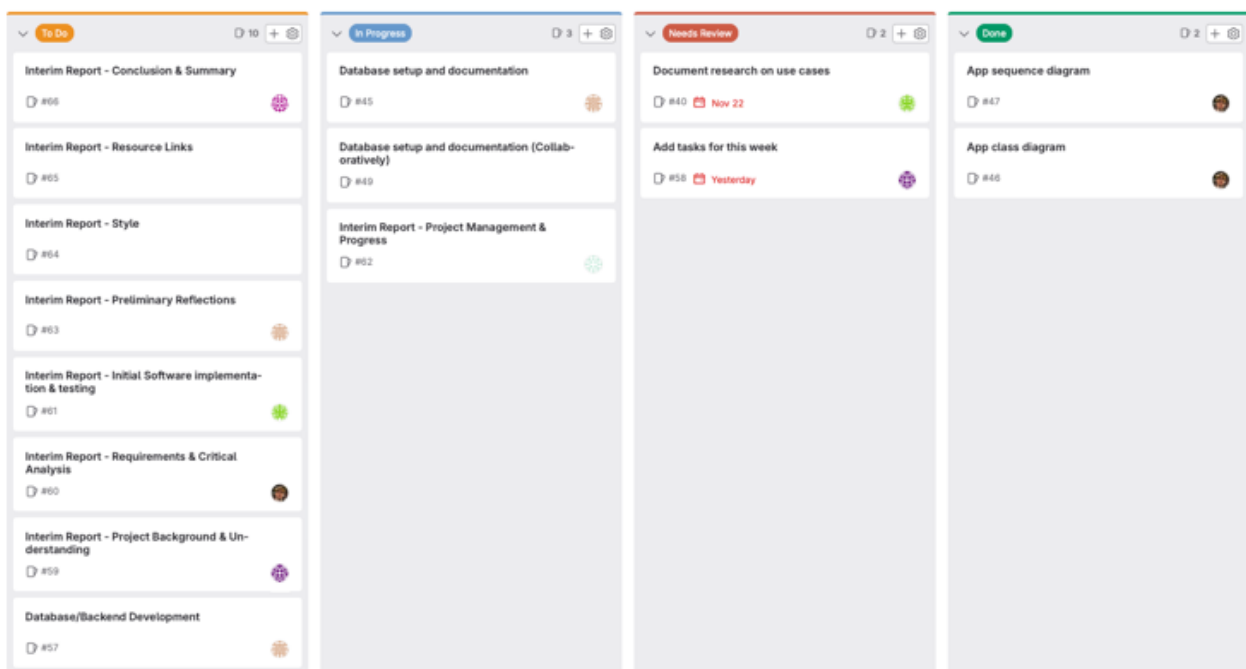
// This component is used to control access to protected routes based on login status
const ProtectedRouteComponent = ({ component: Component }) => {
  // Checking if the user is logged in by retrieving the login status from localStorage
  const isLoggedIn = localStorage.getItem('isLoggedIn');

  return (
    <Routes>
      {/* Conditional rendering based on login status */}
      {isLoggedIn ? (
        // Display the DashboardComponent if the user is logged in
        <Route path="/dashboard" element={ <DashboardComponent /> } />
      ) : (
        // Redirect to the loginComponent if the user is not logged in
        <Route path="/login" element={ <LoginComponent /> } />
      )}
    </Routes>
  );
};

```

Section 4 - Project Management & Progress:

The project's success is due to the agile methodologies the team implemented, as seen through the weekly sprint, tasks assignment and weekly deliverables that they produced. By using Git Issue Boards, they had a clear visual representation of the project's progress, allowing for better organisation and prioritisation. It also meant that they could track ownership of the tasks, ensuring clarity on when tasks need to be done and what they need to do. Also, having multiple Issue Boards made it easier for communication on which tasks have been started, need reviewing or have been completed. Below is an example of what the Git Issue Boards looked like in the middle of Sprint 7:



Furthermore, the team used Microsoft Teams and WhatsApp as their main methods of communication as they believed there were many benefits in doing so. By fully utilising the file sharing aspects of both, it allowed people access to important documentation. By creating multiple different channels in Teams, each individual team have a space where they could communicate without interrupting the overall workflow. Another major benefit was being able to call each other if they had an important or urgent query to discuss was also a major benefit. These communication mediums allowed for a more structured environment for both formal and informal communication.

The weekly sprint retrospectives, clear by the meeting minutes documentation, created by Andreas Sergiou, allowed the team to reflect on what they had done in that sprint and identify areas of improvement, allowing for adjustment of workflow for sprints to come. The weekly cycle of them having meetings, setting tasks, reconvening when those tasks are done and reviewing, was extremely helpful for everyone as a team, as it kept them on track, meeting all objectives. In

terms of meeting frequency, the team would meet up twice every sprint. They would make sure that they had everything that they had discussed and decided in that sprint written down in a meetings minute document for reference. As a team, they are also very efficient in dividing themselves into sub-teams and delegating the tasks that need to be done between members, based on all team member strengths and weaknesses. Here is an example of a Meetings Minute Document, which they used to track everything:

Meeting Minutes Document 14

Date: 29/11/2023

Time: 11:30 AM - 01:45 PM

Location: Computer Science Building, Jubilee Campus, University of Nottingham

Present (in order of attendance):

- Andreas - Team Admin
- Daniel - Deputy Git Master
- Dixon Okoro - Team Member
- Mateusz - Git Master
- Humais - Git Helper
- Martin - Team Leader
- Jacob Stephens - Deputy Team Leader
- Charlotte Fardell - Company Contact

Agenda:

- Welcome and introduction (5 minutes):

- Call the meeting to order
- Converse with present team members
- Briefly introduce the purpose of the meeting

- Progress from last week (25 minutes):

- Here is what every team member has done since the 28th:
 - Andreas - Reviewed and finished the desktop prototype and documentation for it. Also filled the SCRUM and made the meeting minutes document for meeting 14
 - Aaron - Reviewed and finished the desktop prototype and documentation for it.
 - Martin - Finalised the pitch deck for meeting 14
 - Humais - made documentation on the databases - progressed on the database.
 - Daniel - Requirements for data points and did more research on data points for the project - has been in contact with Jacob also
 - Mati - Requirements for data points and did more research on data points for the project - has been in contact with Jacob also
 - Dixon - Tried to do his work, but had issues with it
 - Jacob - Finished all his work on the 28th

- What is discussed in the meeting (45 minutes):

- What the company contact said in the meeting:
 - She said that she was happy with both designs

- Sent over the hexadecimal values for the colour palette.
- She implied that the 'Admin' aspect of the app may be something that we implement after the app has been developed.
 - Only do it if you have enough time.

- Discussion between Andreas and Martin (30 minutes):

- This discussion was to do with a design on the prototypes.
- Martin suggested a new design for the mobile web app and explained why it would be good for use to implement.
- Andreas agreed with him but brought up the fact that we may not have enough time to actually implement this design.
- The discussion was concluded with everyone agreeing to develop the prototype with the initial design created by Andreas and Aaron, and in the next semester, to start using Martin's design.

- Deliverables for next sprint (30 minutes):

- These are the deliverables for the next sprint:
 - Martin - Keep on track with what everyone is doing - helping with the UI development process.
 - Andreas - Start coding the initial Desktop Prototype, one page at a time - discuss with Aaron on what needs to be done - should result in some pages.
 - Aaron - Start coding the initial Desktop Prototype, one page at a time - discuss with Aaron on what needs to be done - should result in some pages.
 - Jacob - Can start work with AI development, work with everyone on that team.
 - Daniel - Can start work with AI development, work with everyone on that team.
 - Humais - Discussed in the meeting on what needs to be done.
 - Mateusz - Can start work with AI development, work with everyone on that team.
 - Dixon - Needs to communicate with Humais on what needs to be done.

Next Meeting: Wednesday 6th December - Time: 11:00PM - Place: Computer Science Building

Submitted by Andreas Sergiou

The team's professionalism, discipline and consistency has been seen throughout this project. They have all demonstrated a strong commitment to their work and have consistently followed the project plan and deadlines, without any deviation from it, not just working as a team but when working individually also. Going a step further, they have implemented a system that allows for everyone to learn from each other and collaborate efficiently. After a team member has finished with their tasks, they would share it with the group, making sure it is done to a high standard. For example, when the AI team were researching on what use case they wanted to focus on, they researched 5, finding data that they could use for each. They then produced in-depth research

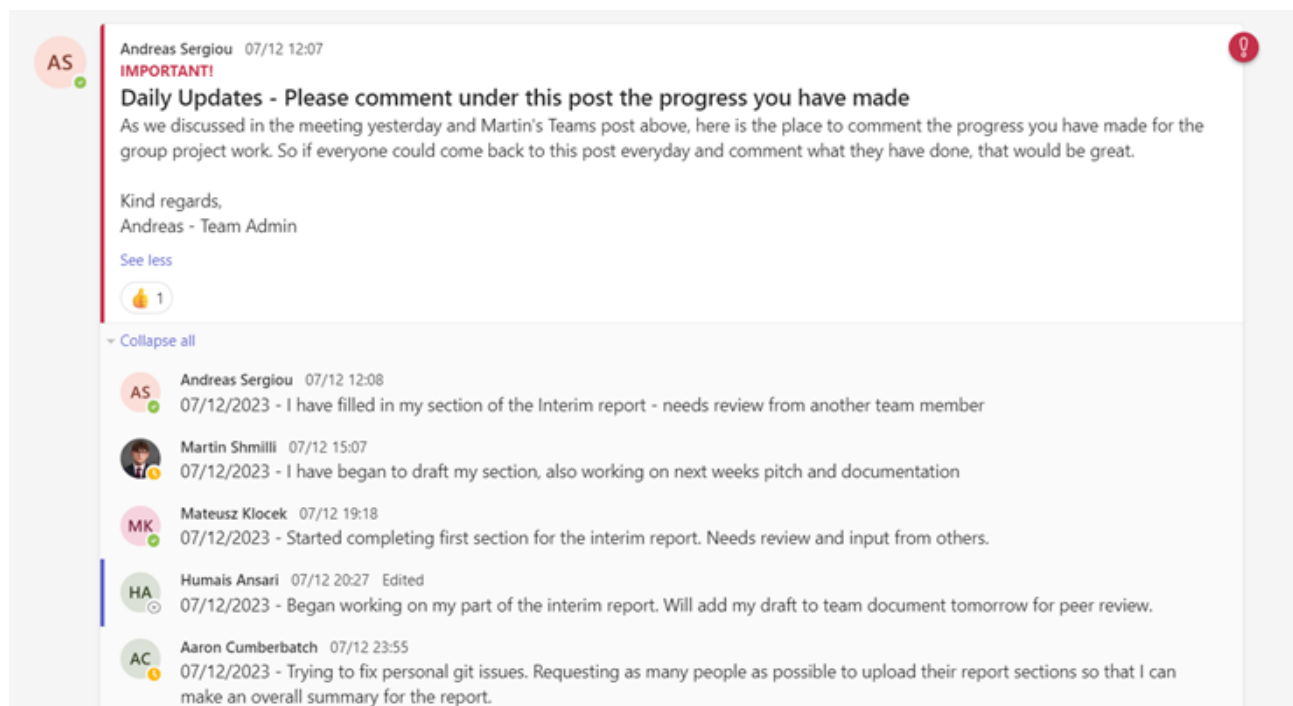
documentation and gave the rest of the team a summary of their findings, fostering cross-learning.

To conclude, the team has a strong commitment to the project, and is progressing smoothly, due to their effective use of project management tools, clear communication channels and regular meetings. They have also shown a professional attitude, showing discipline and self-organised behaviour. While there were some slip-ups in the past sprint, the team have taken this on-board and are committed to addressing and solving these issues in the next sprints. As a result, the team is well-equipped to successfully manage the project and deliver what is required to Eviden.

Section 5 - Preliminary Reflections

The team's primary method of testing was pitching ideas to the industry contact and project supervisor during weekly meetings to ensure that they could test initial concepts and prototypes before committing to a long-term solution.

To showcase the initial designs of the app, Figma was used. Whilst this allowed the front-end sub team to produce vastly appealing prototypes for the industry contact, it also led to some disagreements amongst team members regarding design choices. Fortunately, these issues were handled in a very professional manner as the team members resolved them via meaningful communication. Going forward, the team look to discuss their tasks and ideas more diligently before they are pitched to prevent disagreements from occurring before key deadlines. An actionable approach we have taken recently to improve this is setting up a daily sprint check on Teams to get a concrete idea of everyone's progress.



The team's main method of communication with both the industry contact and project supervisor was through weekly meetings. However, occasionally the team faced issues with setting up a regular recurring time slot for these meetings because of technical issues or because either the supervisor or industry contact was not available. Going forward, they look to create a permanent time slot where all parties can be available for a meeting either on a weekly or biweekly basis. The team's safe and reliable data storage was implemented via the use of MySQL and AWS. Whilst the backend team successfully created a populated database, there is still an opportunity to develop efficient queries for faster data retrieval. This process can be streamlined by developing greater knowledge of SQL statements through online resources. To improve the

relational understanding of the various tables, the team can also look to generate easy to follow diagrams via tools such as <https://drawsql.app/>.

Whilst the AI and backend sub teams have worked effectively to complete their tasks, a potential concern is integrating the solutions and data between the database and AI Models. The team's current workflow as of now has been limited, which will be addressed by developing a more comprehensive architecture that implements compatibility between the data. Therefore, communicational channels between the AI and backend teams should be strengthened to get a better understanding of the requirements and model expectations.

The UI team chose to use React when creating the initial web app. While it is a widely popular library for building user interfaces, it comes with a steep learning curve. It took the UI team a few tries to understand the component-based architecture and to understand and learn the JSX syntax. A practical step the UI team can take is acquiring a centralised source of information to tackle issues faster and lead to a consistent methodology for developing the app. Overall, the team has produced a substantial amount of work since acquiring the project and has received overwhelmingly positive feedback from the project supervisor as well as industry contact.

The team encompasses core strengths of meeting hard deadlines, communicating strongly during pitch meetings, delegating tasks efficiently to correct team members, and precisely choosing the correct software to reach objectives.

But it is important to highlight weak points such as limited integration between sub teams and inconsistent meeting timings. To ensure we can successfully create a functional and desired app, the team will swiftly gather at the start of the next semester to gauge an understanding of the potential workload, assess future tasks, delegate relevant subgroups, and set key objectives to complete.

Section 6 - Conclusion & Summary

In this interim report, the project centred on enhancing sustainable practices through machine learning and advanced analytics. The team has aimed to develop a first working Proof of Concept, driven by the importance of social and environmental sustainability in modern business. The team's vision involved crafting a data-driven sustainable web app using machine learning to forecast photovoltaic power consumption and estimate potential savings. The teams interpreted the brief, set clear milestones, and emphasised efficiency enhancements and cost savings as quantifiable metrics.

The team's collaborative efforts have resulted in comprehensive deliverables covering research, design, and development phases, yielding refined prototypes, robust databases, and evidence of AI model implementation. The overall project management, including weekly sprints and effective communication via platforms such as Microsoft Teams and WhatsApp, ensured steady progress and alignment with aims. Robust data management in Amazon RDS displayed the team's commitment to security and GDPR principles, integrating user control measures within the UI.

Leveraging technologies like Git, React, and Node.js allowed for the project's robustness. This report reflects noteworthy progress the team has made towards their objectives, marked by a structured approach, good team management, and a comprehensive understanding of user requirements. Moving forward, the team will stay dedicated to refining prototypes, optimising systems, and addressing identified challenges to ensure the successful delivery of the final solution of a fully developed mobile and desktop web app to Eviden.